Marshall Davey

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RESEARCH INTERESTS

Mathematical modeling and computer simulation of physical phenomena especially organ scale, fluid-structure interaction cardiovascular models; computational medicine; cardiovascular physiology; computational tool development; high-performance computing; continuum mechanics; computational fluid dynamics; immersed boundary fluid-structure interaction methods; medical devices

WORK EXPERIENCE

Postdoctoral Research Associate June 2024–Present
Griffith Lab, University of North Carolina at Chapel Hill, Departments of Mathematics and Biomedical Engineering
Carolina Center for Interdisciplinary Applied Mathematics Chapel Hill, NC

- Developing a generalizable fluid-structure interaction model platform using an immersed boundary method with finite elements.
- Replicating in vitro systems for mitral and aortic valve flow analysis in silico for method validation and future disease/device investigations.

Graduate Research Assistant March 2019–May 2024
Griffith Lab, University of North Carolina at Chapel Hill, Departments of Mathematics and Biomedical Engineering
Carolina Center for Interdisciplinary Applied Mathematics Chapel Hill, NC

- Developed a novel fluid-structure interaction whole heart model with physiologic three-dimensional descriptions of all four valves able to produce physiologic pressure-volume relations.
- Provided cardiac function analysis by translating numerical pressure, velocity, flow rate, and displacement data into clinical-like metrics.
- Designed model credibility studies for heart model to illustrate dynamic load response via the Frank-Starling mechanism

Graduate Research Assistant August 2018–December 2018
Frolich Lab, University of North Carolina School of Medicine, Neuroscience Center Chapel Hill, NC

- Implemented and joined previously developed cortex and thalamus network models using the Brian 2 Python package
- Tested and investigated neural synchronization from simulated transcranial alternating current stimulation through phase analysis

Research Technician December 2017–August 2018
Elston Lab, University of North Carolina School of Medicine, Department of Pharmacology Chapel Hill, NC

- Utilized cellular level MATLAB model for parameterization and simulation of enzymatic pathways in cystic fibrosis and COPD

EDUCATION

Doctor of Philosophy, Bioinformatics and Computational Biology May 2024
University of North Carolina at Chapel Hill

Master of Physiology, North Carolina State University December 2016

Bachelor of Science, Mathematics, University of North Carolina at Chapel Hill May 2014

AWARDS AND CERTIFICATES

The Langtangen Prize for Outstanding Paper on the Foundations of Numerical Simulation Technology 2025
Ridgeway Scott Foundation (for **M Davey** et al., PNAS Nexus, 2024)

Graduate Certificate in Cardiovascular Science May 2024
University of North Carolina School of Medicine, McAllister Heart Institute

Best Student Poster Award March 2021
Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering Virtual

T32 Predoctoral Training Program July 2019–June 2020
Bioinformatics and Computational Biology, NIH 5T32GM067553-10

PUBLICATIONS

1. M Sadipour, **M Davey**, DR Wells, A Kheradvar, JP Vavalle, and BE Griffith. A whole-heart fluid-structure interaction model for transcatheter mitral valve repair in primary mitral regurgitation. *Annals of Biomedical Engineering*, 2026. DOI: 10.1007/s10439-026-04368-3.
2. **M Davey**, MS Hashemi, GS Pressman, A Kheradvar, and BE Griffith. Patient-specific analysis of ventricular vortex formation and flow energetics in rhematic and calcific mitral stenosis. *Annals of Biomedical Engineering*, 2026. DOI: 10.1007/s10439-026-04364-7.
3. EM Terrell, **M Davey**, DR Wells, and BE Griffith. An effective and accurate semi-implicit time integration scheme for

dynamics in nearly- and fully-incompressible hyperelastic solids. *Journal of Computational Physics*, 564:115125, 2026.

4. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, G Sturgeon, WP Segars, JP Vavalle, CS Peskin, and BE Griffith. Simulating cardiac fluid dynamics in the human heart. *PNAS Nexus*, 3(10):pgae392, 2024.

5. WA Huang, E Negahbani, I Stitt, DJ Passey, S Ahn, **M Davey**, M Dannhauer, T Doan, A Hoover, A Peterchev, S Radtke-Schuller, and F Frohlich. Transcranial alternating current stimulation entrains alpha oscillations by preferential phase synchronization of fast-spiking cortical neurons to stimulation waveform. *Nature Communications*, 12(1):3151, 2021.

CONFERENCE ORAL PRESENTATIONS (PRESENTING AUTHOR)

1. **M Davey**, BE Griffth, A Kheradvar, M Hashemi, P Abdelmaseeh, G Pressman. Fluid-structure interaction modeling of in vitro studies investigating mitral valve disease. *10th International Biofluid Mechanics and Mechanobiology Symposium, Shmuel Einav and Mory Gharib Award Session. Irvine, CA. September 2025.*
2. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Physiologic left heart performance of a fluid-structure interaction model of the human heart. *8th International Conference on Computational and Mathematical Biomedical Engineering. Arlington, VA. June 2024.*
3. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and simulation with a four-chamberd fluid-structure interaction model of the human heart. *2023 Biomedical Engineering Society Annual Meeting. Seattle, WA. October 2023.*
4. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and simulation with a four-chamberd fluid-structure interaction model of the human heart. *2023 Cardiac Physiome Workshop. Irvine, CA. Aprinl 2023.*
5. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and initial experience with a four-chambered fluid-structure interaction model of the heart. *Summer Biomechanics, Bioengineering, and Biotransport Conference. Cambridge, MD. June 2022.*

CONFERENCE POSTER PRESENTATIONS (PRESENTING AUTHOR)

1. **M Davey**, MS Hashemi, GS Pressman, A Kheradvar, BE Griffith. Patient-specific in silico study of ventricular vortex formation and flow energetics in rheumatic and calcific mitral stenosis. *American Heart Association Scientific Sessions. Chicago, Il. November 2026.*
2. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Physiologic left heart performance of a fluid-structure interaction model of the human heart. *FDA/MDIC Symposium on Computational Modeling and Simulation. Hyattsville, MD. April 2024.*
3. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and simulation with a four-chamberd fluid-structure interaction model of the human heart. *IIntegrated Vascular Biology/McAllister Heart Institute Spring Research Symposium. Chapel Hill, NC. March 2023.*
4. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and initial experience with a four-chambered fluid-structure interaction model of the heart. *Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering. Virtual. March 2021*
5. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and initial experience with a four-chambered fluid-structure interaction model of the heart. *Integrated Vascular Biology/McAllister Heart Institute Spring Research Symposium. Virtual. March 2021*
6. **M Davey**, C Puelz, S Rossi, MA Smith, DR Wells, BE Griffith. Construction and initial experience with a four-chambered fluid-structure interaction model of the heart. *Integrated Vascular Biology/McAllister Heart Institute Spring Research Symposium. Chapel Hill, NC. March 2020.*

PROFESSIONAL SOCIETY MEMBERSHIPS

Society for Industrial and Applied Mathematics 2020-Present
American Heart Association 2025-Present

TEACHING EXPERIENCE

How to Learn to Code - Python Summer 2022
Co-instructor
Continuum Mechanics Seminar Spring/Fall 2020
Group Leader
Introduction to Statistical Methods, BCB 720 Fall 2019
Teaching Assistant